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vNAAATS V2 - first-time install guide

For controllers using EuroScope on Windows. Allow about 15 minutes.

> Plugin pack - 1.5b21
>
> Download VatsimNAAATS-plugin.zip
>
> Always the newest build: the plugin DLL, the vNAAATS.asr screen file, an example config and a short README. Your **key** is not in the zip: get it from ganderoceanic.ca/my/vnaaats (certified controllers).

> After installing, read the user manual to see how everything works.

Before you start

- EuroScope installed and working, and a VATSIM controller rating.
- The official **Gander Oceanic sector package from AeroNav** (it also keeps your AIRAC up to date).
- The plugin pack (the download box above) and your **key** from ganderoceanic.ca/my/vnaaats. Do not share the key.

1. Get the Gander package

Download and run the Gander Oceanic package from **AeroNav**: files.aero-nav.com/EXCXO. Open EuroScope with that package once, so you can see the oceanic map, then close EuroScope.

2. Download the plugin

Get the newest zip from the download page (or from the Gander website if you were given a ready-made zip). Unzip it into a folder you will remember, for example **Documents\vNAAATS**. It holds:

- **VatsimNAAATS.dll** - the plugin.
- **vnaaats.cfg** - your settings and key (or **vnaaats.example.cfg**: rename it to **vnaaats.cfg**, open it in Notepad, set **server=https://vnaaats.com** and paste your **key=**).
- **vNAAATS.asr** - the radar screen layout.

Keep the DLL and **vnaaats.cfg** in the **same folder**.

3. Load the plugin in EuroScope

1. Open EuroScope with the Gander package.
2. Other SET > Plug-ins > Load, and pick **VatsimNAAATS.dll**.
3. **Only vNAAATS should be loaded.** Remove any other plugin from that list while you work oceanic. (Discord is fine.)

4. Open the vNAAATS screen

1. Open the file menu and choose **New vNAAATS radar screen**. You get a blank screen with a blue bar at the top.
2. Open the file menu again, choose **Open**, and pick **vNAAATS.asr**.
3. If EuroScope warns about a different sector file, pick your **latest Gander (EXCXO) sector file**.
4. The oceanic map appears. Choose **Save** so you never have to do this again.

If the screen stays blank, check that the Gander package is the one loaded and try the steps again in order.

5. Check it works

- A pop-up shows the plugin version if there is a newer one.
- Open **vNAAATS.log** (next to the DLL). The first line shows the version.
- Open the Flight Data Display in your browser, sign in (name or CID and the password from the same Gander page) and pick your position. Nothing is shown until you have done both.

Updating

Close EuroScope and replace **only VatsimNAAATS.dll**. **Keep your existing vnaaats.cfg** and your saved screen. Do not copy the example config over it or you will lose your key.

Let the display type commands for you (autotype)

The example config has **autotype=1** switched on. The Send buttons for SELCAL, contact me and RCL REQ in the display then type the command into EuroScope for you, and your EuroScope can send the **automatic RCL REQ reminders** (see the manual). If you would rather copy and paste, set **autotype=0** (the automatic reminders then come from another controller's EuroScope).

If something goes wrong

- **EuroScope closes when it starts:** check **server=https://vnaaats.com** (with https) in **vnaaats.cfg**. Install the Microsoft Visual C++ 2015-2022 Redistributable (x86) if the plugin will not load at all.
- **Wrong version in the log:** EuroScope is loading a different copy of the DLL. Other SET > Plug-ins shows which folder it is using.
- **The map is wrong or not oceanic:** you have another sector file loaded. Use the Gander package from AeroNav.
- **No flights appear or changes do not save:** the key is missing, wrong or out of date (keys last 90 days) in **vnaaats.cfg**. Get a new one from ganderoceanic.ca/my/vnaaats.
- **Still stuck:** send **vNAAATS.log** to the Gander Oceanic team.

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vNAAATS V2 - user manual

Every feature of the Flight Data Display (FDD) and the EuroScope plugin, and how to use it. Not installed yet? Start with the install guide.

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1. How it all fits together

vNAAATS has three parts that share one set of flight data:

- **The Flight Data Display (FDD)** - a web page at vnaaats.com/fdd. This is where you work. It lists every oceanic flight as a strip and lets you change flight data, track and transfer aircraft, send messages and run CPDLC.
- **The EuroScope plugin** (VatsimNAAATS.dll) - shows the same aircraft on your radar and in EuroScope's flight plan windows, and carries out what you ask for on the FDD (track, untrack, transfer, SELCAL, contact me).
- **The vNAAATS server** - keeps everything in step, so every controller sees the same level, Mach, track, SELCAL and notes within a few seconds.

> **The FDD is your main screen.** Almost everything can be done from it, and most of what you do in EuroScope can also be done there. Keep EuroScope connected to VATSIM as your oceanic position with the plugin loaded, and use the FDD in a browser window next to it. (EuroScope still does what only it can do: radar, your own tracking and the flight plan windows.)

2. Starting a session

1. Start EuroScope, connect to VATSIM as your oceanic position (for example CZQO_CTR) and check the vNAAATS plugin is loaded.
2. Open vnaaats.com/fdd in your browser.
3. Sign in with the login you were given and choose your **position** (CZQO_CTR, EGGX_CTR and so on). The position is what other controllers see on "last changed" and "tracked by".

Use it as an app: in Chrome or Edge, open the FDD and click the *install* icon at the right end of the address bar (or the three-dot menu, then *Install vNAAATS FDD* / *Apps*). It then opens in its own window with its own icon, which suits a second monitor. It is still the same website, so updates arrive automatically.

Nothing is shown until you have signed in and chosen a position. This protects the data of pilots and controllers: the page stays covered, and the server sends no flight data, until a position has been chosen. There is no view-only mode. Use *change* to pick another position (the page is covered again until you do) and *sign out* when you finish. If your sign-in runs out while you work, the sign-in screen returns.

Keep EuroScope connected while you work: the plugin asks the server every 2 seconds whether you have asked for something (track, transfer, SELCAL and so on), and does nothing if EuroScope is offline.

3. The Flight Data Display screen

- **Left (the big panel):** the filter bar, the controls bar, the natTrak PENDING and CLEARED boxes, and below them the flight strips.
- **Right, top:** the **Flight Plan** panel for the selected aircraft.
- **Right, middle:** natTrak information.
- **Right, bottom:** the **Tracks** panel (today's NATs).
- **Pop-ups:** the aircraft menu (click a callsign), the CPDLC drawer, and the Concorde position reports table.

4. The top bars and filters

Filter bar (top)

- **HL / LL / ALL TK / RR** - high level, low level, all tracks, random routings. The selected one is yellow. **Default** resets the filters.
- **PST** posted flights (cleared through the plugin), **FaD** filed data (known only from the filed flight plan), **WST** westbound only, **TERM** flights with no update for 30 minutes or more. Hover over each box for its description.
- The two boxes on the right are the Zulu clock and the TMI (the day number of the NAT track message).

Controls bar (second row)

- **Your position** - with *change* and *sign out*.
- **Add callsign + Add** - type the callsign of a flight that is on VATSIM and press Add (or Enter). The flight is looked up on VATSIM and stays on the display. Use this for a flight that did not appear by itself, or to bring back one you hid. If it is not found, a yellow message tells you.
- **Hidden (n)** - shown once you have hidden a strip. Pick a callsign from the list to **unhide** it.
- **Search** - type part of a callsign or a fix name. The list shows only the matches; Enter jumps to the first.
- **Show** - *entry -60 / exit +5 min* (the default) shows a flight from 60 minutes before its oceanic entry estimate until 5 minutes after its exit estimate, so the list stays short. *All flights* shows everything. "n hidden" counts the flights the time window is keeping off the list. Flights with no times, the selected flight, CPDLC-connected flights, pending natTrak requests and flights tracked by an oceanic controller always show.
- **Group** - by track, by entry fix, or by flight level (highest first, with the number of aircraft under each heading).
- **Sort** - newest first, entry ETA, callsign, level or last changed.
- **CPDLC button** - opens the CPDLC drawer. It shows how many aircraft are connected, an envelope with the unread count, and flashes "new logon" when a pilot has just logged on. See section 9.
- **n shown** - how many strips are listed.
- **Auto RCL: ON/OFF** - the automatic RCL REQ reminder (section 8). Click to switch it for everyone. Hover to see which controller's EuroScope is the sender.
- **Overdue** (flashing clock) - Concorde flights overdue a position report. See section 11.
- **Warning triangle + number** - how many flights currently have a warning on their strip.

5. Flight strips and their symbols

Each flight is one strip, and strips are grouped under the track or entry fix. The group heading shows the track's fixes, and each strip shows its estimated time over every fix, lined up underneath. Click the **callsign** to open the aircraft menu and select the aircraft (it also becomes the selected aircraft in EuroScope when you use a command on it). It works the other way too: click an aircraft in EuroScope and its strip is outlined in **yellow**, scrolled into view and shown even if a filter, the search box, the time window or Hide would normally keep it off the list (needs plugin 1.5b18 or newer, and the same position on the FDD as in EuroScope).

Every strip shows: callsign, aircraft type, departure and arrival, assigned level, Mach (three digits, so Concorde shows M200), the track, the fix times, and symbols:

- **[CALLSIGN] in brackets** - the pilot is logged on to CPDLC with vNAAATS.
- *** after the callsign** - the aircraft is **not** AGCS-equipped (worked out from the PBN and remarks of the flight plan).
- **Red * after the callsign** - you sent the pilot the natTrak reminder (RCL REQ). It clears itself when the pilot requests clearance or an oceanic controller tracks the aircraft.
- **Pencil** - something on this strip was changed in the last 10 minutes. Hover to see who and when.
- **Warning triangle** - two aircraft are entering on the same track and level within 10 minutes of each other, or the level is not one of the track's flight levels. Hover for the reason.
- **Headset icon** - who the pilot is talking to, taken from the radio frequency they have tuned on VATSIM:
  - Bold green - an oceanic controller
  - Faded green - a domestic controller
  - Orange - UNICOM (122.800)
  - Red, crossed out - online but not on any controller or UNICOM

  Hover to see the callsign and frequency. If VATSIM has no radio information, it shows who is tracking the aircraft in EuroScope instead.
- **Clipboard with a tick** - the clearance was processed on natTrak (hover for the time).
- **Flashing strip** - a natTrak request is waiting, a CPDLC message or logon needs a look, or a Concorde is overdue. Flashing can be turned off (see section 8).

Strips for flights tracked by an oceanic controller (CZQO, EGGX, NAT_FSS and so on) never disappear until the flight has been untracked for 5 minutes. Flights whose pilot has left VATSIM are hidden unless they are selected, on CPDLC, or have a request pending. A filed **NAT track** such as *HOIST NATF GOMUP* is checked against today's tracks and replaced by that track's fixes between the filed entry and exit fix, so the strip sits under the right track. After you send **Contact me** the callsign cell turns **yellow**; if 20 minutes later nobody is

tracking the flight and it is still in oceanic airspace it turns **red**, and it clears once an oceanic controller tracks it. **If EuroScope crashes** or you lose your connection, the plugin tracks your aircraft again about 20 seconds after you are back (within 5 minutes, never an aircraft another controller has taken, and not if someone with a different name has signed in on the FDD at your position). Put *retrack=0* in vnaaats.cfg to turn this off. Waypoints are always shown as four digits (for example 52/50), whatever format the flight plan used.

6. The aircraft menu

Click a callsign. The heading shows the aircraft and the **track** it is on. The tabs along the top are, in order:

- **Track** - check or change the **level**, **Mach** and **SELCAL**, then **Track aircraft** (the button is on the left; the **SELCAL test** button sits under the SELCAL box). The plugin starts tracking the aircraft in EuroScope and the menu shows the result. If another controller is handing the aircraft to you, **Track also accepts that handoff**.
- **Transfer** - pick the controller to hand the aircraft to and press Transfer. Only stations from the CPDLC station list are offered. If none are online it says *No adjacent ATC - Send to UNICOM* (use the UNICOM button in the CPDLC drawer).
- **Release Tag** - releases the aircraft in EuroScope (you are asked to confirm).
- **SELCAL** - enter or change the 4-letter code and use **Send** to run a SELCAL check in EuroScope, **Copy .selcal** to paste it yourself, or **Save code** to store it on the flight. Letters allowed: A-H, J-M, P-S (a red warning names any that are not).
- **CPDLC** - opens the CPDLC drawer for this aircraft if the pilot is connected.
- **Note** - type a short note (up to 15 characters) and press Save note: the EuroScope scratchpad (also in the flight plan panel).
- **Contact Me** - **Send** runs EuroScope's contact-me command for the aircraft. **Copy .contactme** copies it.
- **RCL REQ** - the reminder asking the pilot to request oceanic clearance on natTrak (CPDLC clearance is not available). **Send** types the message into EuroScope for you, **Copy** copies it, **I have sent it - mark with red *** puts the red * on the strip and the radar tag, and **Remove mark** clears it.
- **Hide** - hides the strip **for you only**, in this browser, for 12 hours. Nobody else is affected and EuroScope and VATSIM are not touched. Bring it back from the *Hidden* list or with *Add callsign*.
- **Position reports** - Concorde only. See section 11.
- **Supervisor** (red) - write a message to the supervisors, or press **No Contact** for the standard text, then **Copy .wallop** and paste it in EuroScope.

Track, untrack, transfer, SELCAL and contact me need you to be signed in with a position and EuroScope connected. **Send** buttons need **autotype=1** in vnaaats.cfg (section 13); without it use the **Copy** buttons and paste the command into EuroScope's command line (click in it, Ctrl+V, Enter). EuroScope's contact-me and SELCAL commands work on the *selected* aircraft and take no callsign, so the plugin selects the aircraft for you before it types them.

7. The flight plan panel

Shows the flight you last clicked. The title shows its callsign (in brackets if it is on CPDLC). Boxes you can type in can be changed when you are signed in; the change is saved on the server, every other controller sees it within seconds, and your EuroScope flight plan window updates by itself.

- **ACID, Type, Depart, Dest** - read-only.
- **Entry** - the time at the oceanic entry fix. Edit it and **all the other fix times move by the same amount**.
- **Datalink** - green *CPDLC on* when the pilot is logged on to the vNAAATS CPDLC station, red *No CPDLC* otherwise.
- **ATC** - the callsign of the controller tracking the flight (green), or who the pilot is tuned to on VATSIM: oceanic (green), domestic (pale green), red *UNICOM* or red *NONE*.
- **Level, Mach, Track** - editable. A warning appears if the level is not on the track, or if another aircraft is entering the same track and level within 10 minutes.
- **Route (edit)** - amend the route. Latitude/longitude fixes must be written as two numbers over two numbers (51/20). Forms such as 51N20W are refused with a red warning as you type. Fixes you keep keep their times; new fixes have none. This changes the FDD and server only, **not** the flight plan in EuroScope or on VATSIM.
- **Note** (up to 15 characters) - the EuroScope **scratchpad**. It shows under the tag in EuroScope, goes with the aircraft to the next controller, and anything typed in the scratchpad in EuroScope shows here. Letters, numbers, spaces and / . , + * - only. Clear it to remove the note.
- **SELCAL** - on the route row, editable.
- **Position reports** button - Concorde only.
- **Last change HHMMz by POSITION** - who last changed level, Mach, track, SELCAL or the route.

If the panel says *View only - sign in and pick a position to edit*, nothing can be changed yet: sign in and choose a position.

8. natTrak requests and alerts

Pilots request their oceanic clearance on natTrak. The FDD shows the requests in two boxes under the controls bar:

- **PENDING - needs action** - callsign, NAT, entry fix, level, Mach and ETA. A new request flashes (box row and strip) and plays a sound. Click the row to open natTrak on that track's pending list. **Open natTrak** opens it in a small pop-up.
- **CLEARED** - clearances already issued, with the time. Clicking one selects that flight.

At the top of the CLEARED box, **Visual** turns the flashing and tab-title alert on or off, **Audio** turns the sound on or off, and **Tone** cycles through Ping, Chime and Beep (it plays a preview). Your choices are remembered in your browser.

Automatic RCL REQ reminder

Switched **on by default**. The server watches every flight and, when one has **not** asked for clearance on natTrak, sends it the same RCL REQ message as the aircraft menu button, without anyone clicking. A flight is sent the reminder when:

- its oceanic entry estimate is between **20 and 45 minutes** away (or, if it has no times, it is within **200 nm** of a fix on its route and still more than 20 minutes of flying from it). natTrak does not accept a new request inside 20 minutes of the entry point, and a flight that has already passed its entry point (or been through oceanic airspace) does not need one, so neither is sent a reminder;
- it is online and moving, has no natTrak request, is not tracked by an oceanic controller, and has not been reminded already (by hand or automatically).

The message is typed into **EuroScope** by one oceanic controller who is online with the plugin and **autotype=1** (the first one alphabetically, if there are several). It is sent one flight at a time, and the strip gets the red * when it has gone. If it fails (for example autotype is off) it tries again after 5 minutes, up to three times. Your EuroScope command line is used for a second or two when it sends, so expect a short interruption if you are typing. Use **Auto RCL: ON** in the controls bar to switch it off for everyone; the setting is remembered by the server.

9. CPDLC

Pilots connect to the vNAAATS CPDLC station and you talk to them from the CPDLC drawer, opened with the **CPDLC** button in the controls bar.

Signing in

Each controller signs in with their **own Hoppie logon code**. The code is typed into the drawer, kept in the server's memory only while you are signed in, and never saved. Your position sets the station (CZQO = CZQX, EGGX = EGGX, NAT_FSS = NATX). Use **sign out of CPDLC** when you leave. Restarting the server logs everyone out of CPDLC, so sign in again if that happens.

Working with pilots

- **Connected aircraft** are listed at the top of the drawer. Click one to see the message thread.
- **New logon:** when a pilot logs on, the alert sound plays **once**, their callsign flashes in the drawer and on the strip, and the CPDLC button flashes "new logon". It stops when you look at that aircraft.
- **New message:** a sound plays, the button shows an envelope with the unread count, and the strip flashes until you open it.
- **Replying:** quick buttons *unable*, *standby*, *roger*; the *reply...* choices (Wilco/Unable, Affirm/Negative, Roger, None) set the response type; or write your own text and press **Send**.
- **Standard messages...** - the TopSky message list.
- **Transfer** - when you hand an aircraft that is on CPDLC to another controller, the pilot is sent *CONTACT <station> ON <frequency>* and is logged off when they answer WILCO or ROGER.
- **UNICOM** (asks you to confirm) - sends "THERE IS NO FURTHER ATC AVAILABLE. SERVICE TERMINATED. MONITOR ADVISORY ON 122.800", logs the pilot off when they answer WILCO or ROGER, and releases the tag in EuroScope.

You need a controller position to send messages. Aircraft on CPDLC show their callsign in brackets, and the flight plan panel shows *CPDLC on*.

10. Tracks panel

Bottom right. The heading shows the TMI. Westbound and eastbound tracks are listed with their validity time once on the heading (for example *Westbound 1130-1900z*). Each track is two lines: its name and fixes, then its flight levels. The same fixes appear as the column headings over the strips.

11. Concorde position reports

For aircraft type CONC, a **Position reports** button appears in the aircraft menu and flight plan panel. It opens a table with every fix on the route and three times: **Calc** (calculated), **Est** and **Actual**. Type times as four digits, or press **now** to fill the current time. Reported times replace the calculated times on the strip.

A Concorde **flashes** (strip, button and table row), and the top bar shows a red *overdue* count, when the next fix's time passed more than 3 minutes ago with no report.

12. What you see in EuroScope

- **Aircraft icons** on the radar point along the heading. **Filled** means cleared, **outline** means not cleared yet.
- **[CALLSIGN]** in brackets on the tag - the pilot is on CPDLC.
- *** under the callsign** - not AGCS equipped.
- **Red *** - the natTrak reminder was sent from the FDD.
- **Note under the tag** - the scratchpad. You can type it in EuroScope or on the FDD.
- **Conflict alerts** flash for aircraft that are too close. A one-level difference is allowed 200 ft of tolerance, so aircraft exactly one level apart do not flash.
- **Flight plan window:** the track, level, Mach and SELCAL boxes are the same data as the FDD. The Datalink box shows *CPDLC ON* or *CPDLC OFF*. The SELCAL box accepts four letters such as ASCM or AS-CM; clear it to go back to the code in the remarks. The **FDD** button in the menu bar opens the display.
- **Clear / Unclear** - clearing an aircraft that is oceanic sets transponder code **2000** automatically (switch it off with **squawk=0**).
- **Handoff** - when you hand an aircraft that is on CPDLC to another controller (Co-ordination / Active Authorities > Transfer), the pilot is sent the contact message and logged off after WILCO/ROGER, as with the FDD Transfer.
- **Updates:** a pop-up appears when a newer plugin is available and opens the download page. Replace **only the DLL** and keep your vnaaats.cfg.

Anything you change in EuroScope shows on the FDD within seconds, and anything you change on the FDD appears in EuroScope in the same way.

13. Settings in vnaaats.cfg

A text file next to VatsimNAAATS.dll. Open it in Notepad, save, then restart EuroScope.

- **server=https://vnaaats.com** - the server (keep the https).
- **key=...** - your access key from the Gander Oceanic members page. Keep it private.
- **squawk=1** - give code 2000 on oceanic clearance (0 = off).
- **autotype=1** - lets the FDD **Send** buttons type into EuroScope for you. The plugin brings EuroScope to the front, clears the command line and types one command at a time. With **autotype=0**, or if Windows refuses, the command is copied for you to paste instead and the menu tells you so.

14. A typical shift, step by step

1. Connect EuroScope to VATSIM as your position. Open the FDD, sign in and choose the same position.

2. Check the **Tracks** panel for today's NATs and the **Show** setting for the strips you want.
3. An aircraft approaching the oceanic boundary has not asked for clearance: open its menu and use **RCL REQ > Send**, then **I have sent it - mark with red ***.
4. The pilot requests clearance on natTrak: the PENDING box flashes and plays a sound. Click the row to open natTrak and issue the clearance. The flight moves to CLEARED.
5. Check level, Mach, track and SELCAL in the **flight plan panel**. Fix anything wrong; add a **Note** if you want one under the tag.
6. When the aircraft is yours: **Track** tab > **Track aircraft** (or accept the handoff the same way).
7. A pilot logs on to CPDLC: the sound plays, the callsign and button flash. Open the drawer, reply or send the standard messages you need.
8. Aircraft leaving your area: **Transfer** tab > pick the next controller. With no one online, use **UNICOM** in the CPDLC drawer.
9. A flight you do not want to see: **Hide** it. It only disappears for you, and comes back from the *Hidden* list.

15. Problems and questions

- **The page looks old or a new feature is missing:** hard refresh (Ctrl+Shift+R on Windows, Cmd+Shift+R on Mac). The header shows the display and server versions.
- **"No data on remote":** the server is not answering. Try again in a minute and tell the Gander Oceanic team if it stays.
- **Track, transfer, SELCAL or contact me does nothing:** check you are signed in with a position, EuroScope is connected as that position, and the plugin is loaded. The result (or the reason) is shown at the bottom of the aircraft menu.
- **Send buttons only copy: autotype=1** is missing from vnaaats.cfg or Windows refused to let the plugin type. Use the Copy buttons.
- **A flight is missing:** it may be outside the time window (see **Show**), hidden by you (see **Hidden**), or filtered. Use **Add callsign** to bring it in.
- **No sound:** browsers only play sound after you have clicked somewhere on the page. Click once, and check **Audio** is ON in the natTrak box.
- **Changes not saving:** you have not chosen a position, or the key in vnaaats.cfg is missing or wrong.
- **Reporting a bug:** include the callsign, what you clicked, what you expected, what happened, the plugin version (first line of vNAAATS.log) and the log file. Report at GitHub issues or to the Gander Oceanic team.

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